

Chp ~ 7

Alternating Current

⇒ Alternating Voltage and Current developed

An Alternating current (or voltage) is one whose magnitude changes continuously with time and whose direction reverses periodically

when a closed coil is rotated rapidly in a strong magnetic field. the magnetic flux changes periodically.

$$\phi = BA \cos \theta$$

$$\frac{d\phi}{dt} = -BA \sin \theta \times \frac{d\theta}{dt}$$

$$= -BA \omega \sin \theta$$

Now, $V = -N \frac{d\phi}{dt}$

$$= -N \times -BA \omega \sin \theta$$

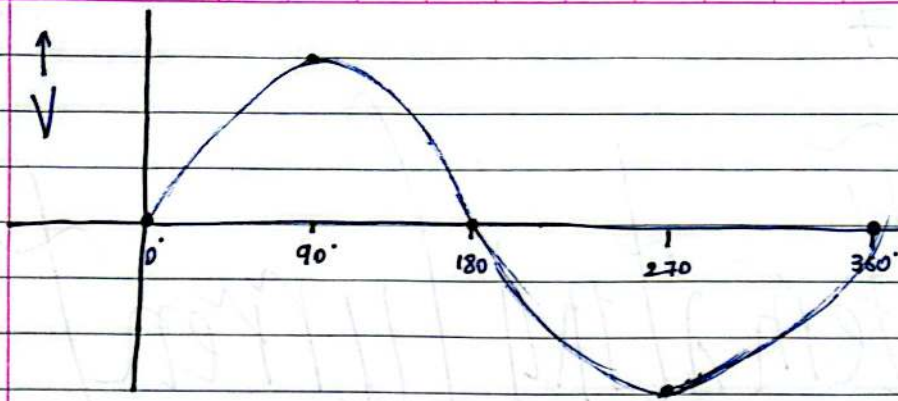
$$V_0 = NBA \omega \sin \theta \Rightarrow V = V_0 \sin \theta$$

$$V = V_0 \sin \omega t$$

$$\theta = 0^\circ \Rightarrow V = 0$$

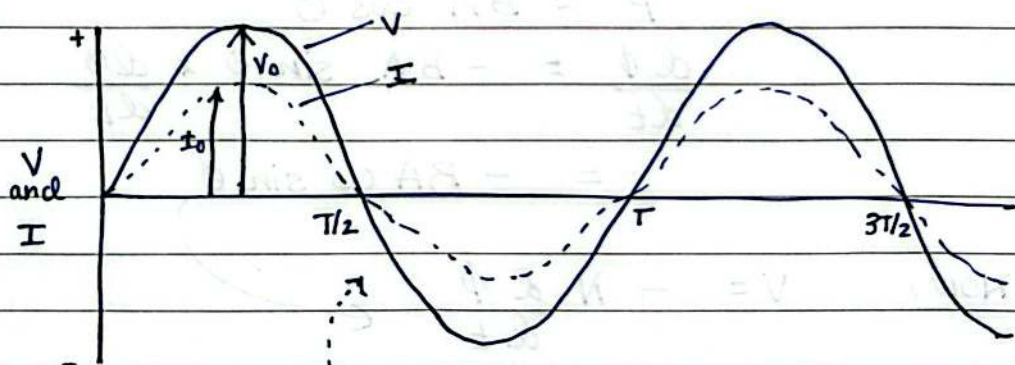
$$\theta = 90^\circ \Rightarrow V = NBA \omega$$

max



Similarly, $I = I_0 \sin \omega t$ Sinusoidal Curve.
 max current

- Peak value (V_0, I_0): the maximum value attained by voltage/current
- Time period (T): time for one complete cycle.
- Frequency ($f = 1/T$): number of cycles per second.
- Angular frequency (ω) = $2\pi f = 2\pi/T$



Here, I will always be lower than voltage as Resistance acts upon it and ($V = IR$).

⇒ Mean (or Average) value of Alternating Current.

i) For one cycle:

$$\begin{aligned}
 I &= I_0 \sin \omega t \\
 I_m &= \frac{\int_0^T I \cdot dt}{\int_0^T dt} \\
 &= \frac{I_0 \int_0^T \sin \omega t \, dt}{[t]_0^T} \\
 &= \frac{I_0}{\omega} \frac{[-\cos \omega t]_0^T}{T} \\
 &= \frac{I_0}{\omega T} [(-\cos \omega T) - (-\cos 0)] \\
 &= \frac{I_0}{\omega T} [-\cos 2\pi + \cos 0] \\
 &= \frac{I_0}{\omega T} [1 - 1] = \frac{I_0}{\omega T} \times 0 = 0
 \end{aligned}$$

Thus, for a complete cycle the average or mean value of Alternating current is zero.

ii) For half Cycle :

$$I = I_0 \sin \omega t$$

$$I_m = \frac{\int_0^{T/2} I dt}{\int_0^{T/2} dt}$$

$$= \frac{I_0 \int_0^{T/2} \sin \omega t}{\left[t \right]_0^{T/2}}$$

$$= \frac{I_0 \left[-\cos \omega t \right]_0^{T/2}}{\omega \left[T/2 - 0 \right]}$$

$$= \frac{2 I_0}{\omega T} \left[\left(-\cos \frac{\omega T}{2} \right) - \left(-\cos 0 \right) \right]$$

$$= \frac{2 I_0}{2\pi} \left[\left(-\cos \frac{2\pi}{2} \right) + \cos 0 \right]$$

$$= \frac{I_0}{\pi} [+1 + 1]$$

$$= I_0 \times \frac{2}{\pi}$$

$$\boxed{I_m = I_0 \times 0.637}$$

Thus, the mean or average value of AC for a half cycle ($t=0$ to $T/2$) is 0.637 times or 63.7% of peak cycle.

⇒ Root mean square value of alternating current.

$$I = I_0 \sin \omega t$$

$$I_{rms}^2 = \frac{\int_0^T I^2 dt}{\int_0^T dt}$$

$$= \frac{\int_0^T I_0^2 \sin^2 \omega t dt}{T}$$

$$= \frac{I_0^2}{T} \int_0^T \sin^2 \omega t dt \quad \left[\sin^2 \theta = \frac{1 - \cos 2\theta}{2} \right]$$

$$= \frac{I_0^2}{T} \int_0^T \frac{1 - \cos 2\omega t}{2} dt$$

$$= \frac{I_0^2}{2T} \left[t - \frac{\sin 2\omega t}{2\omega} \right]_0^T$$

$$= \frac{I_0^2}{2T} \left[\left(T - \frac{\sin 2\omega T}{2\omega} \right) - (0 - 0) \right]$$

$$= \frac{I_0^2}{2T} \left[\left(T - \frac{\sin 2\omega T}{2\omega} \right) - (0 - 0) \right]$$

$$= \frac{I_0^2}{2T} [T - 0]$$

$$I_{rms} = \frac{I_0}{\sqrt{2}}$$

$$= I_0 \times 0.707$$

⇒ Different types of A.C Circuits. (L - C - R)
 inductor capacitor resistor

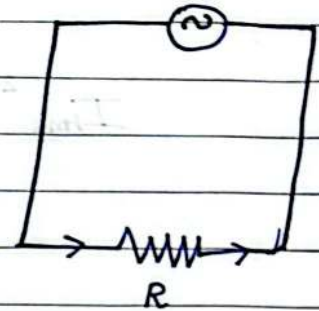
i) Circuit Containing Resistance only :

$$V = V_0 \sin \omega t$$

$$V = IR$$

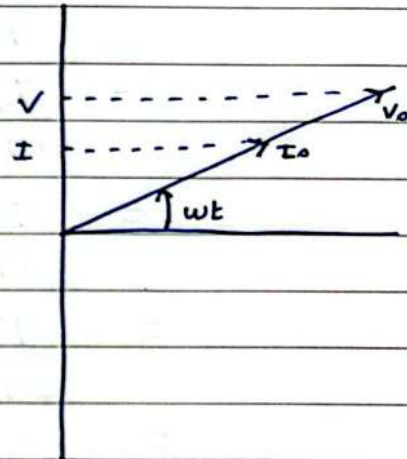
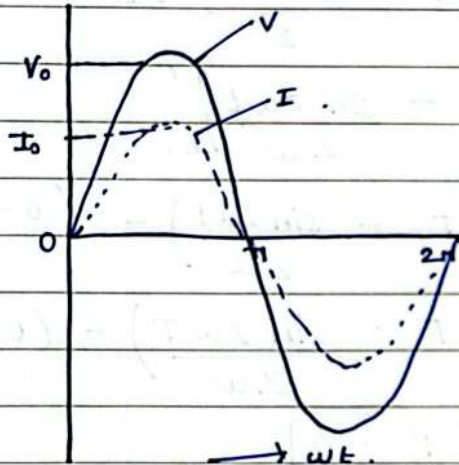
$$V_0 \sin \omega t = IR$$

$$I = (V_0 / R) \sin \omega t$$



$$\therefore I = I_0 \sin \omega t$$

→ In a pure resistor the current I is always in phase with the applied voltage V .



a) Phase relation through graph

b) Phasor diagram.

ii) Circuit containing Inductance only:

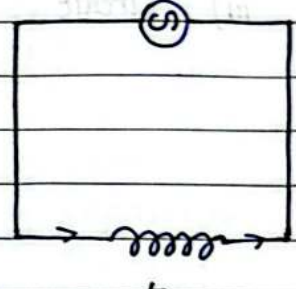
$$V = V_0 \sin \omega t$$

$$V = V_0 \sin \omega t$$

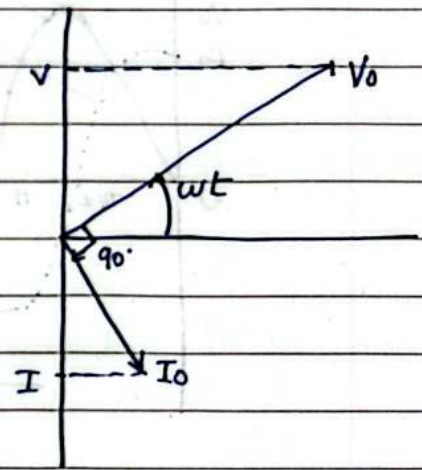
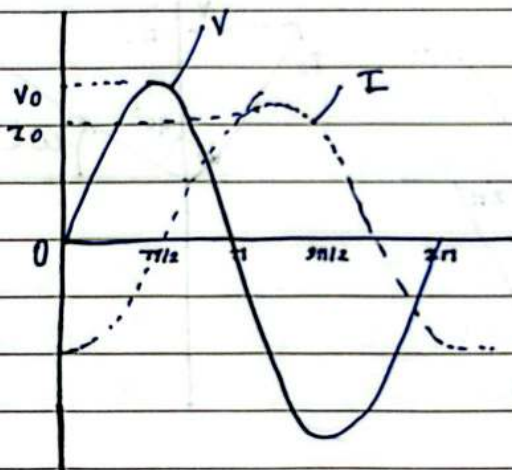
$$I = I_0 \sin \left[\omega t - \frac{\pi}{2} \right]$$

$$\therefore I = \frac{V_0}{\omega L} \sin \left[\omega t - \frac{\pi}{2} \right]$$

$$\boxed{\frac{V_0}{\omega L} = I_0}$$



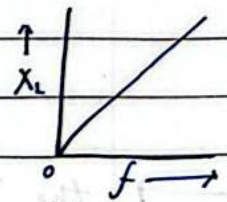
$\therefore$ In a pure Inductor, the current lags behind the voltage by a phase angle of $\pi/2$ or 90° (or the voltage leads the current by a phase angle of $\pi/2$ or 90°).



$\Rightarrow$ Inductive Reactance: It represents the 'effective opposition' of the coil to the flow of alternating current. Known as inductive reactance (X_L)

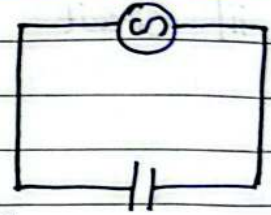
$$\boxed{X_L = \omega L = 2\pi f L}$$

$\rightarrow X_L \propto f$, directly proportional



$\rightarrow X_L = 0$ (for D.C) as frequency is zero for D.C

iii) Circuit containing Capacitance only:



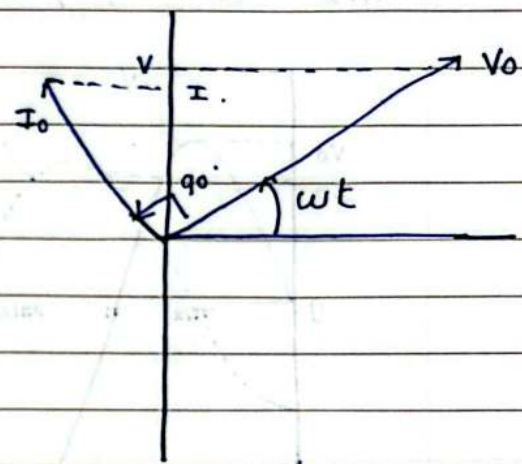
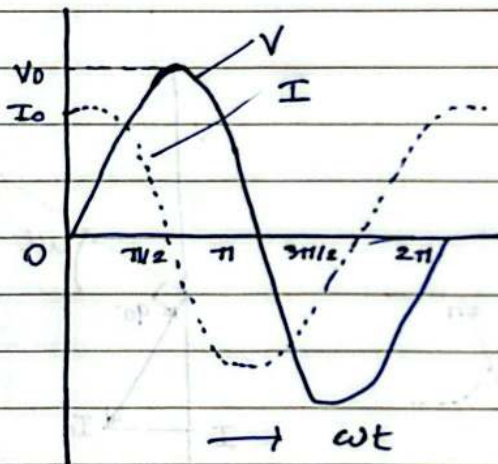
$$V = V_0 \sin \omega t$$

$$I = \frac{V_0}{1/\omega C} \sin \left[\omega t + \frac{\pi}{2} \right]$$

$$I = I_0 \sin \left[\omega t + \frac{\pi}{2} \right]$$

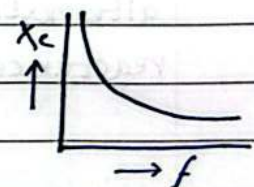
$$I_0 = \frac{V_0}{1/\omega C}$$

$\therefore$ In a perfect capacitor, the current leads the voltage by a phase angle of $\pi/2$ or 90° (or the voltage lags behind the current by $\pi/2$ or 90°)



$\Rightarrow$ Capacitive Reactance: It represents the 'effective opposition' of the capacitor to the flow of alternating current. (X_c)

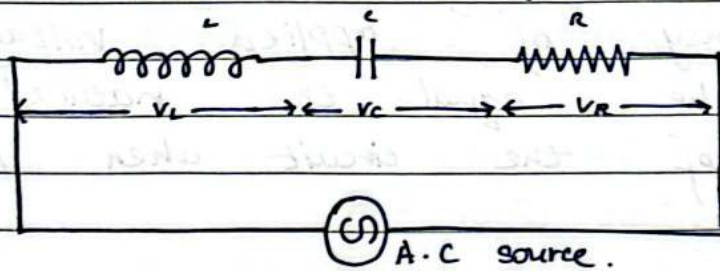
$$X_c = \frac{1}{\omega C} = \frac{1}{2\pi f C}$$



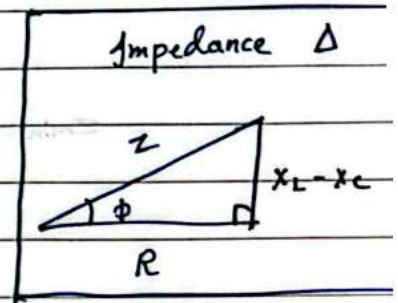
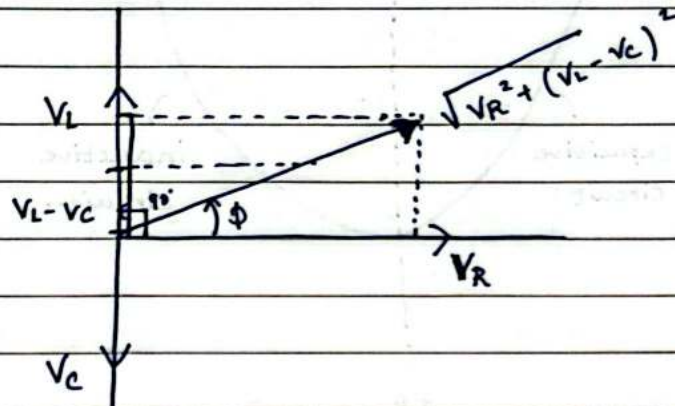
$\rightarrow$ the $X_c \propto 1/f$ (inversely proportional)

$\rightarrow X_c = \infty$ (for D.C) as frequency = 0. ($\frac{1}{0} = \infty$)

iv] Circuit containing Inductance, Capacitance, Resistance in series : (L-C-R circuit).



L - Inductor
 C - Capacitor
 R - Resistor.



The effective resistance = $\sqrt{R^2 + (X_L - X_C)^2}$ is called the 'impedance' (Z) of the circuit.

- 1) $X_L > X_C \Rightarrow I$ lags V by ϕ
- 2) $X_L < X_C \Rightarrow I$ leads V by ϕ
- 3) $X_L = X_C \Rightarrow \text{Resonance} = \omega L = \frac{1}{\omega C}$

$$\omega^2 = \frac{1}{LC}$$

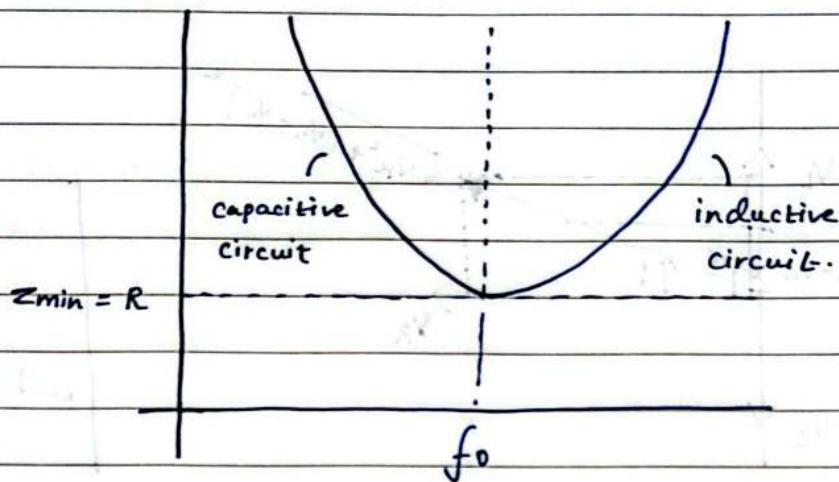
Resonant frequency $\longrightarrow$

$$2\pi f = \frac{1}{\sqrt{LC}} \text{ or } f = \frac{1}{2\pi\sqrt{LC}}$$

$\Rightarrow$ Power factor $(\cos \phi = \frac{R}{Z})$

* When $X_L = X_C$, then $Z = R$
 (Z will be minimum.

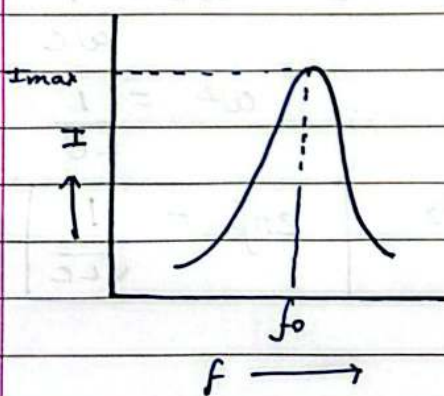
Thus, the condition for resonance is that the frequency of applied voltage should be equal to natural frequency of the circuit when resistance is zero.



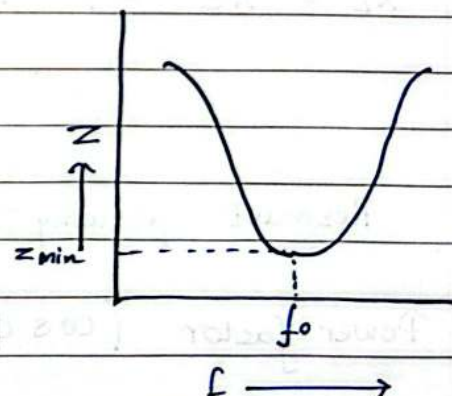
$\Rightarrow$ Resonant circuits.

$$f = \frac{\omega}{2\pi} = \frac{1}{2\pi\sqrt{LC}}$$

Relation between frequency.



a) Current



b) Impedance.

⇒ Power in A.C Circuits.

i] Circuit containing pure resistance only :

$$P = VI$$

$$P = V_0 \sin \omega t \times I_0 \sin \omega t$$

$$P = V_0 I_0 \sin^2 \omega t$$

for one complete cycle $\Rightarrow \sin^2 \omega t = 1/2$.

$$P = V_0 I_0 \times 1/2$$

$$P = \frac{V_0}{\sqrt{2}} \times \frac{I_0}{\sqrt{2}}$$

$$\boxed{P = V_{rms} \times I_{rms}}$$

ii] Circuit containing both Inductance and Resistance :

V leads I by ϕ

$$V = V_0 \sin \omega t$$

$$I = I_0 \sin (\omega t - \phi)$$

Now, $P = VI$

$$P = V_0 \sin \omega t \times I_0 \sin (\omega t - \phi)$$

$$P = V_0 \sin \omega t \times I_0 [\sin \omega t \cos \phi - \cos \omega t \sin \phi]$$

$$P = V_0 I_0 [\sin^2 \omega t \cos \phi - \frac{1}{2} \sin 2\omega t \sin \phi]$$

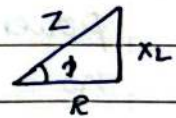
$$P = V_0 I_0 [\sin^2 \omega t \cos \phi - \frac{1}{2} \sin 2\omega t \sin \phi]$$

for one complete cycle

$$P = V_0 I_0 \left[\frac{\cos \phi}{2} \right] = \frac{V_0}{\sqrt{2}} \frac{I_0}{\sqrt{2}} \cos \phi$$

$$\boxed{P = V_{rms} \times I_{rms} \times \cos \phi}$$

$$= V_{rms} \times I_{rms} \times \frac{R}{\sqrt{R^2 + (\omega L)^2}}$$



$$\tan \phi = \frac{X_L}{R}$$

$$\cos \phi = \frac{R}{Z}$$

$$Z = \sqrt{R^2 + X_L^2}$$

⇒ Wattless Current

If a circuit contains pure inductor or pure capacitor alone (resistance is zero).
 ($\phi = 90^\circ$)

$$P = V_{rms} \times I_{rms} \times \cos 90^\circ$$

$$\boxed{P = 0}$$

When power is zero but current is flowing. This current is called wattless current.

Thus, if Resistance in an A.C circuit is zero, although current flows in the circuit, yet the power remains zero, that there is no energy dissipation in the circuit.

⇒ Choke Coil.

It is a coil of high inductance and very low resistance used to control A.C current in a circuit without significant power loss.

While a resistor would waste power as a heat.

Analytical basis for why chokes are preferred over resistors for controlling A.C current economically.

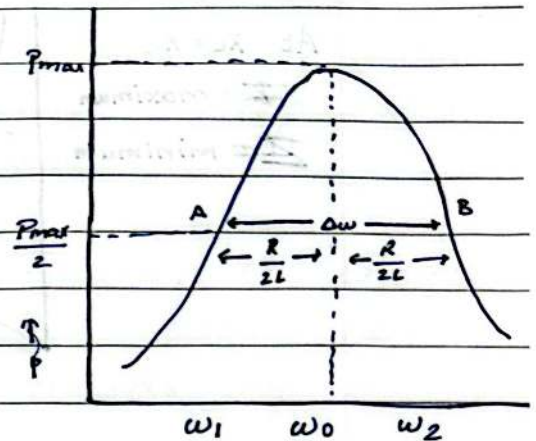
⇒ Half Power points, Bandwidth, Q-Factor.

✓. large
Derivation

$$P_{\max} = \frac{1}{2} \frac{V_0^2}{R}$$

At resonance,

$$P = \frac{P_{\max}}{2}$$



Formula (Qualitative) for Bandwidth.

$$\Delta \omega = \frac{R}{L}$$

(Q-factor)

$$Q = \frac{1}{R} \sqrt{\frac{L}{C}}$$

At resonance ($\omega = \omega_0$) $Z = R$ (minimum).

so current is at its peak and power is at its maximum.

half power points are the two frequencies (ω_1) below (ω_0) and (ω_2) above (ω_0).

At which power is half of its max value.

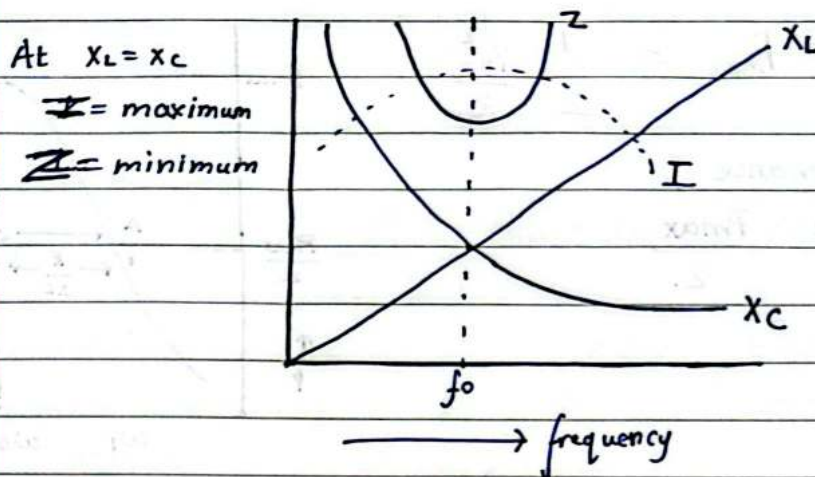
⇒ Electrical Oscillations in L-C circuit.

frequency of electrical oscillations where ($R=0$).

ideal circuit →
$$f = \frac{1}{2\pi} \sqrt{\frac{1}{LC}} = \frac{1}{2\pi} \sqrt{\frac{1/C}{L}}$$

actual circuit → damped oscillatory current is when the oscillations stop after some time.

⇒ Dependence of X_L , X_C , Z and I on frequency.



⇒ Alternating Current Generator (A.C Dynamo).
 It converts mechanical energy to electrical energy

→ It works on principle of electromagnetic induction.

when a closed coil is rotated rapidly in a strong magnetic field the number of magnetic flux lines passing through the coil changes continuously. Hence an emf is induced and current flows in it in direction by F.R.R.

→ Constructions.

1. Armature : A coil of many turns of copper wire wound on a soft iron core (to increase flux linkage).
2. Field Magnets : Strong magnets that provide uniform magnetic field.
3. Slip rings : two rings fixed to rotating shaft connected to two ends of armature.
4. Brushes : Carbon contacts that press against slip rings conducting current to ext circuit.

⇒ Transformers.

→ Based on principle of mutual Induction. used for converting large alternating current at low voltage into small current at high voltage and vice versa.

When the current in a primary coil changes the changing flux linked with a neighbouring secondary coil induces emf in it.

Note: A laminated core is used in the coils to reduce the energy loss due to eddy currents, increasing its efficiency.

→ For an ideal transformer $P_{\text{input}} = P_{\text{output}}$

$$\frac{I_p}{I_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p} = r \quad \text{transformation ratio.}$$

Energy losses in a transformer.

- i) Copper losses : Copper in coil heats up and wastes energy
- ii) Eddy current loss : Eddy currents are set up in iron core and generates heat.
- iii) Flux losses : Coupling between P and S is never perfect, thus not whole magnetic flux is passed.

⇒ Utility of A.C in comparison to D.C

- i) Cost of transmission for A.C is less than D.C
- ii) A.C equipments are more durable for use than D.C (to convert AC → DC rectifiers)
DC → AC Inverters
- iii) A.C can be controlled by choke coil while D.C needs resistance.